

Drug treatments to protect your bones

There are different drug treatments that can help prevent or treat bone loss and reduce the risk of fractures.

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About treatments for bone health

Your doctor may suggest you have treatment to protect your [bone health](#). This may be because:

- [bone health tests](#) show you have low bone density or a high risk of fracture
- you are having a cancer treatment that increases your risk of developing osteopenia or [osteoporosis](#) and risk of bone fracture.

If your GP or cancer doctor thinks you need specialist advice for osteoporosis, they may refer you to a specialist hospital team. These teams may include doctors, nurses and physiotherapists who work closely with occupational therapists, and pain specialists.

Calcium and vitamin D supplements

Your doctor may advise you to take calcium and vitamin D supplements. This may be because you:

- are not getting enough calcium or vitamin D
- are having a treatment that can increase your risk of [osteoporosis](#).

They may do blood tests to check your calcium levels. Or they may organise a [fracture risk assessment](#) or [DXA scan](#) before deciding whether you need to take supplements.

If you need supplements, your doctor may prescribe them for you.

You can buy vitamin D supplements without a prescription. If you are thinking about buying calcium supplements, always talk to your cancer doctor, pharmacist or GP. They can talk to you about whether you need to take them and what dose you should take. You may be able to get enough [calcium](#) from your diet. We have more information about [keeping your bones healthy](#).

Your GP or cancer doctor may measure your blood calcium levels regularly if you are having [bisphosphonate](#) treatment. You may also be prescribed calcium and vitamin D supplements as part of this treatment.

Bisphosphonates

[Bisphosphonates](#) are a widely used bone strengthening treatment. They can help to:

- prevent bone loss caused by hormonal therapy and other cancer treatments
- strengthen bones and reduce the risk of fractures in people with osteoporosis
- prevent cancer spreading to the bone
- treat cancer that has spread to the bone ([secondary bone cancer](#) or bone metastases).

How do bisphosphonates work?

Bisphosphonates reduce the activity of cells that break down bone (osteoclasts). This slows down bone loss. Bisphosphonates do not replace lost bone, but they can stop further bone loss. This helps to make the bone stronger.

Bisphosphonates used to treat osteoporosis

There are many bisphosphonates used to treat osteoporosis. These include:

- alendronic acid alendronate (Fosamax[®]) given as a tablet
- [ibandronic acid](#) or ibandronate (Bonviva[®]) given as a tablet or an injection
- risedronate sodium (Actonel[®]) given as a tablet
- [zoledronic acid](#) or zoledronate (Aclasta[®]) given via a drip into the vein (intravenous).

How to take bisphosphonates

Bisphosphonates used to prevent bone thinning and treat osteoporosis are usually taken as tablets or capsules. You may take them daily, weekly or sometimes monthly.

You take the tablets first thing in the morning on an empty stomach. Take them with a glass of water. You need to sit or stand upright for 30 to 60 minutes after taking the tablet. This helps the drug move quickly into the stomach and reduces the risk of it staying in the throat or gullet (oesophagus), where it can cause irritation. It also helps stop the drug coming back up from the stomach into the gullet. This can happen when you are lying down or bending.

Some people with [osteoporosis](#) may be given bisphosphonates by a drip (infusion) into a vein (intravenously). It can usually be given in the outpatient department at the hospital.

Side effects of bisphosphonates

Possible side effects include:

- indigestion – this is more likely when taking bisphosphonate as a tablet
- a sore throat or inflamed gullet (the tube that goes from the mouth to the stomach) – this is more likely when taking bisphosphonate as a tablet
- pains in your muscles and joints
- flu-like symptoms, which usually settle after the first dose.

Talk to your cancer doctor, specialist nurse, pharmacist or GP if you have any of these side effects.

Tell them before taking any more of this drug, if:

- swallowing is painful or difficult
- you have indigestion that is new or getting worse.

Rare side effects

Thigh bone fractures (atypical fractures)

A small number of people taking bisphosphonates develop fractures in their thigh bone without any obvious cause. Sometimes both thigh bones are affected.

If you have any pain in your thigh, hip or groin, tell your doctor and explain that you are taking bisphosphonates. They can arrange tests to check the thigh bones for any signs of weakness or fracture.

Osteonecrosis of the jaw (ONJ)

A rare side effect of bisphosphonate treatment is osteonecrosis of the jaw (ONJ). It happens when healthy bone tissue in the jaw becomes damaged and dies. This can cause loose teeth and problems with the way gums heal.

The risk of ONJ is very low if you are taking bisphosphonates to treat osteoporosis. ONJ is more likely to happen when they are used to treat cancer that has spread to the bones (secondary bone cancer) and you are having other cancer treatments. This is because higher doses of bisphosphonates are used to treat secondary bone cancer. It is also more likely if you have poor dental health.

Before starting bisphosphonate treatment, your doctor will advise you to see your dentist for a check-up. Any dental treatment needs to be done before starting bisphosphonates. This includes making sure any dentures fit well. During treatment, continue to have regular dental check-ups and look after your teeth and gums well.

Avoid having dental treatment that could affect your jawbone during bisphosphonate treatment. If you need dental treatment, tell your doctor before you have the treatment. This includes having dental implants put in, or having a tooth or root removed, except in an emergency. It is fine to have fillings, gum treatments or a scale and polish.

Denosumab (Prolia®)

A drug called [denosumab](#) may be used to strengthen and protect bones from the effect of cancer or cancer treatment.

Denosumab is given once every 6 months as an injection just under the skin. This is called a subcutaneous injection. If you have denosumab, your doctor may advise you to take calcium and vitamin D supplements.

Denosumab can cause some side effects. These include:

- skin problems, urine and chest infections
- constipation
- pain in the arms or legs
- a rash.

Rarely, denosumab may also cause osteonecrosis of the jaw (ONJ). Osteonecrosis of the jaw is when healthy bone tissue becomes damaged and dies. It may also rarely cause thigh bone fractures that happen with minimal or no trauma.

You would also need to have a dental check-up and any dental work done before starting denosumab. Let your doctor know if you need any dental work after starting denosumab.

Raloxifene (Evista®)

[Raloxifene](#) is used to treat osteoporosis in people who have been through menopause. It is only used if you cannot take bisphosphonates. You take raloxifene daily as a tablet.

Raloxifene acts in a similar way to the hormone oestrogen. It can:

- reduce the breakdown of bone
- keep the bones strong
- reduce the risk of fractures.

You should not take raloxifene if you have, or have had, breast cancer. This is because raloxifene may make the treatments used less effective. Your doctor can tell you if it can be used after treatment for breast cancer has been completed.

Hormone replacement therapy (HRT)

There are different types of hormone replacement therapy. They can be given as tablets, injections, gels or skin patches.

Oestrogen replacement

Oestrogen helps protect your bones. If your oestrogen levels are low because of cancer treatment, your doctor may give you HRT to replace the oestrogen. This will depend on the type of cancer you have. HRT can help to protect bone health and reduce [menopausal symptoms](#).

HRT may contain oestrogen and progesterone. This is called combined HRT. Or it may contain oestrogen only. It is often given as tablets or skin patches (transdermal patches).

HRT can have benefits for bone health. But there are also possible side effects or risks. The risks and benefits of HRT depend on:

- the type of cancer you have had – it may not be used if you have or have had breast or ovarian cancer
- your risk of cancer of the ovary and breast
- the type of HRT (combined or oestrogen only)
- whether HRT is taken as tablets or skin patches
- your age
- your general health
- whether you have a family history of blood clots.

HRT is not suitable for everyone. Your doctor will tell you if it is right for you. If you are thinking about having HRT, it is important to discuss the possible benefits and risks with your doctor. This will help you decide if HRT is right for you.

Testosterone replacement

If you have a low level of testosterone because of cancer treatment, you can have testosterone replacement therapy to get back to a normal level. This helps increase bone density. If you have or have had prostate cancer, testosterone therapy may not be suitable for you. Your cancer doctor can talk to you about this.

We have more information if you are [transgender](#) and are taking gender-affirming hormones.

Treatment with parathyroid hormone (PTH)

The parathyroid glands make parathyroid hormone (PTH). These glands are attached to the thyroid gland in the front of the neck. PTH helps the body absorb calcium and stimulates the body to make bone.

A type of PTH called teriparatide (Forsteo®) is made as a treatment in a laboratory. You inject it just under the skin (a subcutaneous injection). A nurse can teach you how to do this yourself. You have the injection every day for up to 24 months (2 years).

People who have broken bones because of severe osteoporosis may be given teriparatide. It cannot be given to people with cancer in the bone. It is also not suitable for people who have had radiotherapy to an area of bone.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our bone health and cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Coleman R, et al. European Society for Medical Oncology. Clinical Practice Guidelines – Bone health in cancer: ESMO Clinical Practice Guidelines. Ann Oncol. 2020. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/supportive-and-palliative-care/bone-health-in-cancer-patients> [accessed April 2024].

European Society for Medical Oncology. Patient Guide on Bone Health. Cancer can affect bone health in several ways such as by spreading to the bones (bone metastases) or cancer treatment causing bone loss, which makes the bones more fragile and likely to fracture. 2022. Available at: <https://www.esmo.org/for-patients/patient-guides/bone-health-in-cancer> [accessed April 2024].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help